ESP-Drone Setup Guide

From a fresh Windows PC to a flashed drone streaming live data to cfclient

This guide sets up the **Robocode frozen version** of ESP-Drone. Every part is pinned to an exact, tested version, so it works the same today and next year. Parts A and B are done **once per PC**; everything you need for them is in one folder of the code, `robocode-setup`.

Part	What you do	Time
A	Download the Robocode ESP-Drone code	5 min
B	Double-click **SETUP.bat**: it installs Git, ESP-IDF 5.0.7, Python 3.11 and cfclient	20-40 min
C	Build the firmware and flash it to the drone	10 min
D	Connect to the drone and check live data in cfclient	5 min
E	Troubleshooting	as needed

What you need

Item	Notes
Windows 10 or 11 PC	With internet (setup downloads about 1 GB) and permission to install software.
ESP-Drone (ESP32-S2 or ESP32-S3)	The Robocode drones are ESP32-S2. Battery charged.
USB cable	Must be a **data** cable (some phone-charging cables are power-only).
Access to the code	A GitHub account with access to **RobocodeUK/esp-drone-idf5**, or a copy of the folder from your teacher.
Gamepad (optional)	For flying from cfclient, e.g. a PS3/PS4/Xbox USB controller.

What is in the robocode-setup folder

File	What it is
SETUP.bat	Double-click this. Installs everything below that the PC is missing.
ROBOCODE_SETUP_GUIDE.pdf	This guide.
install_cfclient.bat	Installs or repairs just cfclient (SETUP.bat runs it for you).
DOWNLOAD_LINKS.txt	The exact installer links and their fingerprints, if you ever need them by hand.
downloads	Created by SETUP.bat. Holds the downloaded installers; can be deleted afterwards.

Part A - Download the code (once per PC)

Use **one** of these two options. Both end with the code in `C:\esp\esp-drone`.

1 **Option 1 - Download ZIP (no extra software needed)**

- Go to **github.com/RobocodeUK/esp-drone-idf5** and sign in.
- Click the green **Code** button, then **Download ZIP**.
- Create the folder `C:\esp`.
- Right-click the downloaded ZIP, choose **Extract All**, and extract it into `C:\esp`.
- Rename the new folder `esp-drone-idf5-main` to `esp-drone`.

2 **Option 2 - Git clone (if Git is already installed)**

Open **Command Prompt** and run:

```
mkdir C:\esp
git clone https://github.com/RobocodeUK/esp-drone-idf5.git C:\esp\esp-drone
```

CHECK This file must exist: `C:\esp\esp-drone\robocode-setup\SETUP.bat`. If it is one folder deeper (e.g. `esp-drone\esp-drone-idf5-main\...`), move the files up so the path matches. Keep this exact short path: long paths or paths with spaces cause build errors on Windows.

Part B - Run SETUP.bat (once per PC)

1 **Start the setup**

In File Explorer open `C:\esp\esp-drone\robocode-setup` and double-click **SETUP.bat**. Read the first screen and press any key to start. It works through three stages and skips anything the PC already has:

- **[1/3] Git for Windows** - downloaded and installed automatically.
- **[2/3] ESP-IDF 5.0.7** - downloads the 916 MB installer, then opens its window (see step 2).
- **[3/3] Python 3.11 and cfclient** - installed automatically, plus a **cfclient (ESP-Drone)** shortcut on the desktop.

NOTE When Windows asks *Do you want to allow this app to make changes?*, click **Yes**. Every download is checked against a fingerprint before it runs; if a file does not match, SETUP.bat deletes it and stops.

2 **Click through the ESP-IDF installer window**

When the ESP-IDF installer opens:

- Keep the default install folder `C:\Espressif`.
- If it offers to **apply fixes** (e.g. long path support), accept.
- Leave the **drivers** ticked (Silicon Labs CP210x, FTDI, Espressif USB) and the target chips ticked. The drivers let Windows talk to the drone over USB.
- Click **Install**, wait 15-30 minutes, then click **Finish**. SETUP.bat carries on by itself.

3 Wait for "Setup complete"

> **CHECK** The SETUP.bat window ends with **Setup complete**. You now have an **ESP-IDF 5.0 CMD** shortcut in the Start menu and a **cfclient (ESP-Drone)** shortcut on the desktop. You can delete the `downloads` folder to free about 1 GB.

> **NOTE** If it says **Setup FAILED**, read the message above it, fix the problem (often the internet connection, or clicking No on a permission prompt) and run SETUP.bat again. Finished stages are skipped.

Part C - Build and flash the firmware

> **IMPORTANT** Always build and flash from the **ESP-IDF 5.0 CMD** shortcut in the Start menu. A normal Command Prompt or PowerShell window does not have the tools set up and will show errors.

1 Plug in the drone and find its COM port

Connect the drone to the PC with the USB data cable. Open **Device Manager** (right-click Start) and expand **Ports (COM & LPT)**. Look for **Silicon Labs CP210x USB to UART Bridge (COM3)**. The number (here **COM3**) is your port.

> **NOTE** If no new port appears: try another USB cable (it may be power-only) or another USB socket. Unplug and replug the drone and watch which entry appears.

2 Open ESP-IDF 5.0 CMD and go to the code folder

```
cd C:\esp\esp-drone
```

3 Check which chip the drone has (ESP32-S2 or ESP32-S3)

Replace COM3 with your port:

```
python -m esptool -p COM3 chip_id
```

Look for the line `Chip is ESP32-S2` or `Chip is ESP32-S3`.

4 Select the chip type

For ESP32-S2 (the Robocode drones):

```
idf.py set-target esp32s2
```

For ESP32-S3 use `idf.py set-target esp32s3` instead. You only need to do this again if you switch to a drone with the other chip.

5 Build, flash and watch the drone start up

```
idf.py -p COM3 flash monitor
```

The first build takes 5-10 minutes; later builds are much faster. When flashing finishes, the drone restarts and its log scrolls in the window.

> **CHECK** In the log you should see `MPU6050 I2C connection [OK]` and, after the drone has been kept still for a few seconds, `STAB: Ready to fly.` Lines saying VL53L1X, Z-down or PMW3901 **[FAIL]** are normal: those are optional add-on sensors that are not fitted.

Press **Ctrl +]** to leave the monitor. The drone is now flashed.

Part D - Connect to the drone

> **SAFETY** For first tests, **remove the propellers** or hold the drone firmly. When a gamepad is connected, moving the thrust stick spins the motors.

1 Power on the drone and keep it still

Connect the battery (USB is not needed for flying). Put the drone on a flat surface and do not touch it for about 10 seconds while it calibrates its sensors.

2 Connect the PC to the drone's Wi-Fi

In the Windows Wi-Fi list, choose **ESP-DRONE_XXXXXXXXXXXX** (each drone has its own number) and enter the password **12345678**. Windows will say *No internet*; that is normal.

> **IMPORTANT** Windows may switch back to your normal Wi-Fi after about 40 seconds because the drone has no internet, which cuts off cfclient. To stop this, go to **Settings > Network & internet > Wi-Fi > Manage known networks**, click your normal network and untick **Connect automatically when in range**. Turn it back on when you finish.

3 Open cfclient and connect

If you are using a gamepad, plug it in **before** opening cfclient. Start **cfclient (ESP-Drone)** from the desktop. Click **Scan**, choose `udp://192.168.43.42:2390` in the list next to it, then click **Connect**.

> **CHECK** On the **Flight Control** tab, roll, pitch and yaw update live. Tilt the drone gently by hand and the numbers and horizon should follow it, and keep following it for minutes. The drone's green LED blinks quickly while connected.

4 Set up a gamepad (optional)

Menu **Input device**: select your controller, then choose a mapping (for a PS3/PS4 pad, **PS3_Mode_1** is a good start). Use **Input device > Configure device mapping** to change it. Keep the thrust stick fully down when connecting.

5 Finish

Click **Disconnect** in cfclient, unplug the drone battery, and reconnect the PC to your normal Wi-Fi.

Part E - Troubleshooting

Problem	Fix
SETUP.bat: *download failed* or *does not match the expected fingerprint*	Check the internet connection and run SETUP.bat again (downloads resume). If the fingerprint still does not match, do not install the file by hand; tell your teacher.
SETUP.bat: *Git did not install* or *ESP-IDF 5.0.7 was not found*	You probably clicked **No** on a permission prompt, or changed the ESP-IDF install folder. Run SETUP.bat again, click **Yes**, and keep `C:\Espressif`.
`'idf.py' is not recognized`, or errors like *tool xtensa... has no installed versions* / *python_env ... doesn't exist*	You are in a normal Command Prompt. Open **ESP-IDF 5.0 CMD** from the Start menu and try again.
No COM port appears for the drone	Use a different (data) USB cable or USB socket. If it still does not appear, run the ESP-IDF installer from the `downloads` folder again and make sure the **drivers** are ticked.
Flashing fails: *Failed to connect* or *Access is denied*	Close anything else using the port (another monitor window, Arduino IDE). Check the COM number. If it still fails, hold the **BOOT** button on the drone board while the flash starts.
Drone log shows `MPU6050 I2C connection [FAIL]`. cfclient connects but the data never moves	The motion sensor has locked up (common after a crash or on a low battery). Unplug **both** the USB cable and the battery, wait 10 seconds, then power on again. Charge the battery.
cfclient data freezes after about 40 seconds	Windows switched back to your normal Wi-Fi. See the **IMPORTANT** box in Part D step 2.
cfclient data freezes after only a few seconds, or cfclient will not start	Repair it: double-click **install_cfclient.bat** in the robocode-setup folder.
cfclient: *Socket is already connected* or will not reconnect	Close cfclient completely and open it again.
Scan / Connect does nothing	Check the PC is connected to the drone's Wi-Fi (not your normal network) and the drone is powered.
Build errors mentioning long paths or missing files	Make sure the code is at exactly `C:\esp\esp-drone` (no spaces).

For teachers: keeping the setup frozen

Everything is pinned so nothing changes unless Robocode changes it:

Part	Pinned to
Firmware	**RobocodeUK/esp-drone-idf5** (based on Espressif esp-drone commit db0f6562, the last version for ESP-IDF 5.0). esp-now 2.5.2 is included in the code, so nothing is downloaded at build time.
Installers	Git 2.47.1, ESP-IDF 5.0.7 offline, Python 3.11.9 - official links with SHA-256 fingerprints in `robocode-setup\DOWNLOAD_LINKS.txt`
cfclient	**RobocodeUK/crazyflie-clients-python**, commit 76d5093
cflib (drone link)	**RobocodeUK/crazyflie-lib-python**, commit f2baeab (adds the Wi-Fi keep-alive fix)
Python libraries	Exact versions in `robocode-setup\requirements-cfclient.txt`

• Do not update to Espressif's latest esp-drone: it needs ESP-IDF 5.5 or 6.1.
• To get Robocode updates later, download the ZIP again (or run `git pull`), then run **install_cfclient.bat**.
• More details are in `ROBOCODE_README.md` in the code folder.